MIL-S-38725A
3 February 1967
SUPERSEDING
MIL-S-38725
13 May 1965

MILITARY SPECIFICATION

SCRUBBING MACHINE, PAVEMENT
SELF-PROPELLED, RIDER OPERATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers two types of self-propelled, pavement scrubbing machines with rotary scrubbing brushes and rubber squeegee.

1.2 Classification. Scrubbing machines shall be of the type specified (see 6.2).

Type I - Minimum scrubbing capacity of 56,000 square feet per hour (sq. ft./hr.)
Type II-Minimum scrubbing capacity of 76,000 sq. ft./hr.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproofed Kraft
UU-T-81	Tags, Shipping and Stock
MMM-A-260	Adhesive, Water Resistant, (For Sealing Waterproofed Paper)
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof-For Packaging and Sealing

Military

MIL-P-116	Preservation, Methods of

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MIL-P-514	Plates: Identification, Transportation-Data and Blank
MIL-T-3351	Tractor, Full Tracked, Low-Speed; Tractor, Wheeled, Agriculture; and Tractor, Wheeled, Industrial: And Their Attachments, Packaging of

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION (ICC)

Motor Carrier Safety Regulations

(Application for copies should be addressed to the Interstate Commerce Commission, Bureau of Motor Carriers, Washington, D.C. 20423.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

J551	Measurement of Vehicle Radio Interference (30 to 400 Megacycles).
J552	Radio Noise Suppressors.

(Information as to the availability of this handbook may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York City, New York 10017.)

3. REQUIREMENTS

3.1 Preproduction model. This specification provides for preproduction model inspection.

3.2 Design. The scrubber shall consist primarily of a tank to dispense clean water, a tank(s) to receive the returned dirty water, a vacuum pick up system,

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a series of rotating brushes, a squeegee, and a control system mounted on a self-propelled chassis.

3.2.1 Safety devices. The scrubbing machine shall be fitted with safety devices or covering for all parts that present safety hazards. The devices shall include guards for belts, pulleys, chains, gears, guide ways, and sprockets. All guards shall be designed to provide access to the guarded parts. When special safety devices are required, they shall be as specified (see 6.2).

3.2.2 Lubrication. All bearings (except sealed-for-life-type), mating gears, and sliding parts shall be provided with means for lubrication. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.2.3 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or scrubbing machine.

3.2.4 Material. All parts used in the fabrication of the scrubbing machine shall be new and of the quality necessary to produce a scrubbing machine to meet the requirements described herein.

3.3 Construction. The scrubbing machine shall be complete so that it can be used for any operation for which it was designed. Construction shall be free from any characteristics or defects that will prevent the machine from passing any of the examination(s) and tests in section 4.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the scrubbing machine.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Painting. Unless otherwise specified (see 6.2), the scrubbing machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.4 Components. Components shall be the same as those normally furnished on the manufacturer's standard product provided they meet the requirements of this specification. All components shall be constructed to withstand normal use. All controls shall be capable of being easily operated from the operator's normal operating position.

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3.4.1 Chassis. The chassis shall consist of a commercially available farm, industrial, or warehouse type tractor, or shall be of the manufacturer's standard design. The chassis shall be of a size and weight that shall accomodate all of the mounted components and shall have provisions for a riding operator. The weight of all components, all tanks filled (fuel and water) as normally used, and an operator, shall not exceed 80 percent of the chassis rated load capacity.

3.4.1.1 Engine. The engine shall be the manufacturer's standard commercial model provided it meets the requirements of this specification. It shall be a four stroke cycle, gasoline type and shall be equipped with an exhaust spark or flame arrester. It shall have sufficient horsepower for efficient operation of the scrubbing machine under severe conditions without exceeding 80 percent of the engine rated speed. Motive and scrubbing power shall be supplied by the same engine.

3.4.1.2 Fuel tank. The fuel tank shall have sufficient capacity to operate the engine at rated-load operation for not less than 8 hours. It shall be designed to withstand the shock and vibration loads imposed by the scrubbing operation.

3.4.1.3 Electrical system. The electrical system shall be the manufacturer's standard commercial system except as specified herein.

3.4.1.4 Starting system. The engine shall be started by an electrically energized starting motor actuated from the instrument panel or toeboard.

3.4.1.5 Radio interference suppression. The scrubbing machine shall be suppressed for radio interference in accordance with SAE Standard J551. Suppressors used shall conform to SAE Recommended Practice J552.

3.4.2 Water tanks. One clean water tank of not less than 100 gallon capacity for the Type I scrubber and not less than 150 gallon capacity for the Type II scrubber shall be provided. Tank(s) for returned dirty water shall be provided with each type of scrubber. The returned dirty water tank(s) shall have a capacity of at least 115 gallons for Type I scrubbers and at least 170 gallons for Type II scrubbers. Provisions for flushing all tanks shall be provided.

3.4.3 Scrubbing head. A scrubbing head consisting of multiple rotary brushes, sized and arranged so as to provide clean scrubbing in a 58 to 72 inch swath, shall be provided. The brushes shall be mounted in a manner which will provide self leveling. Brush drive may be either chain and sprocket; gear, shaft and U joint; pneumatic; or hydraulic, provided spare parts are readily obtainable from commercial sources.

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3.4.3.1 Brushes. Each machine shall be furnished with a set of brushes for scrubbing. They shall be of circular shape with either one piece or sectional backing blocks of either laminated or solid construction. The machine shall be so constructed that the brushes, or sections thereof, may be readily replaced without the use of special tools. The diameter at the extremety of the bristles, measured parallel to the brush back shall be not more than 2-1/2 inches larger than the diameter of the brush back. The diameter of the brush bristles shall be measured at the free end of the tufts with no weight applied on the brush. Arrangement and spacing of the tuft holes and the quantity of tufts provided, shall be such that the normal flare of the brush will, with the exception of the center hole, afford a continuous and effective polishing or scrubbing area of fiber.

3.4.3.1.1 Plate. Each brush shall have a metal back plate. The plate shall be adaptable to the machine model upon which it is to be used. This plate shall afford a secure and efficient attachment of the brush to the machine and shall also permit easy and convenient detachment. The plate shall be capable of being easily removed for use on other brushes.

3.4.3.1.2 Tufts. Brush tufts shall be made of 100 percent palmetto fiber. The tufts shall be inserted in holes in the brush block of not less than 1/4 inch nor more than 5/16 inch in diameter and shall be securely wired or held in place by any other suitable means so as to withstand a pull on each tuft of 80 pounds, minimum. The holes for the tufts shall be drilled on 1/2 inch centers and symmetrically arranged covering the brush block. The tufts shall have a free length of not less than 1-1/2 inches when measured at right angles to the surface of the brush block. The free ends of the tufts shall form a uniform surface parallel to the surface of the brush block so that the machine will operate with no apparent wobble. The peripheral tufts in the brush block shall have a suitable spread or flare which will permit brush contact with the floor up to a vertical surface.

3.4.3.2 Scrubbing head control. The scrubbing head shall be equipped with hydraulic lift and pressure controls. The controls shall provide positive scrubbing pressures ranging from brush weight to 200 pounds in excess of brush weight.

3.4.4 Vacuum pickup. The scrubber shall be equipped with a vacuum system to pick up dirty water or sludge and deposit it in the dirty water return tank(s).

3.4.5 Squeegee. A squeegee constructed of gum rubber, neoprene, or other suitable material shall be provided and shall be mounted behind the scrubbing head and vacuum pickup. The squeegee shall be not less than 4 inches longer than the scrub swath. If a curved or overlapping "V" squeegee is provided, the total squeegee shall be provided with easily operable lift and pressure controls. The squeegee shall leave no residue while scrubbing machine is turning at its shortest turning radius. Components of the squeegee shall be available from commercial sources.

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3.4.6 Hydraulic system. The hydraulic system shall include all necessary pumps, valves, piping, cylinders, and pressure controls. A means of determining the fluid level in the hydraulic reservoir shall be provided. An overload protection shall be provided on all high pressure lines to prevent damage to system components.

3.4.7 Skirts. Skirts shall be provided if they are furnished as either a standard or an optional item. The skirts shall be constructed of a wear-resistant, flexible material and shall allow scrubbing to curbs without being damaged.

3.5 Size and capacities. Sizes and capacities of types I and II scrubbing machines shall conform to the requirements shown in Tables I and II, respectively.

Table I

a.	Width overall, inches	(not more than)	72
		(not less than)	58
b.	Length overall, inches	(not more than)	125
		(not less than)	105
c.	Height overall, inches	(not more than)	72
		(not less than)	56
d.	Ground clearance, inches	(not less than)	3

Table II

a.	Width overall, inches	(not more than)	84
		(not less than)	70
b.	Length overall, inches	(not more than)	146
		(not less than)	126
c.	Height overall, inches	(not more than)	77
		(not less than)	61
d.	Ground clearance, inches	(not less than)	3

3.5.1 Scrubbing capacity. The scrubbing machine shall have minimum scrubbing capacity of not less than 56,000 sq. ft./hr. for Type I or 76,000 sq. ft./hr. for Type II. The top scrubbing speed shall be not less than 2 miles per hour.

3.6 Nameplate. Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each scrubbing machine:

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Manufacturer's Name
Manufacturer's Model Number
Manufacturer's Serial Number
Contract Number
Date of Manufacture
US

3.7 Identification plate. Unless otherwise specified (see 6.2), all machines supplied shall carry a separate identification plate of corrosion resistant metal. Identification plates shall conform to MIL-P-514, Type I, Style 2.

3.8 Workmanship. Workmanship of the scrubbing machine and accessories shall be of the quality prevailing among manufacturers normally producing scrubbing machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection

b. Quality conformance inspection

4.2.1 Preproduction inspection. Preproduction inspection shall be applied to the first scrubbing machine produced under the terms of the Contract or Purchase Order. Failure of the preproduction model to pass the examination or one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), preproduction inspection shall consist of the examination in 4.3 and all tests in 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being prepared for delivery under terms of the contract or purchase order. Failure of the scrubbing machine to pass the examination or one or more tests shall be cause for rejection. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the following:

a. Examination (4.3)

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b. Functional test of components to demonstrate proper adjustments, etc.

c. Inspection of preservation, packaging, packing,and marking for shipment and storage (4.5).

4.3 Examination. The scrubbing machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests

4.4.1 Operational test. The preproduction model shall demonstrate its capability by scrubbing a concrete surface of not less than 18,750 sq. ft. upon which deposits of spilled crankcase oils, accumulated grease and other fuel and lubricant spillage are evident. All free running or freshly spilled oils, lubricants or fuels shall be completely removed from the surface. Accumulated grease or oil deposits not in excess of 1/8 inch thickness shall be completely removed from the surface. Accumulated deposits in excess of 1/8 inch thickness may be presoaked to facilitate removal. Scrubbing, vacuuming, and squeegeeing shall be accomplished in sufficient time to assure fulfillment of capacity requirements in 3.5.1.

4.4.1.1 Reliability. After the operational test specified in 4.4.1, the amount of brush wear shall be determined. The brush's useful life expectancy under operating conditions specified in 4.4.1 shall be demonstrated to be not less than 250 hours.

4.4.2 Cleaning swath and effectiveness test. The scrubbing machine shall be tested to determine if the scrubbing head conforms to the requirements of 3.4.3.

4.4.3 Brush lift test. The scrubbing head control shall be tested to determine conformance with 3.4.3.2.

4.4.4 Scrubbing speed test. The scrubbing machine shall be operated to determine conformance to the speed requirements of 3.5.1.

4.4.5 Vacuum pickup test. The vacuum pickup system shall demonstrate its capability to pick up and return to the dirty water tank all free running water and sludge except that entrapped in cracks more than 1/4 inch deep.

4.4.6 Squeegee test. The squeegee shall be tested to determine conformance to 3.4.5.

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4.4.7 Radio interference suppression test. The scrubbing machine shall be tested to determine conformance to SAE Standard J-551. Suppressors shall be examined to determine conformance to SAE recommended practice J-552.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing,and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 General

5.1.1 Materials. Materials shall be as specified herein and in applicable specifications. References to "P-type" preservatives are to those listed in MIL-P-116.

5.1.2 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage, pilferage, and loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins, washers and other attaching parts removed shall be reinstalled in one of the mating parts and secured to prevent their loss. Valve, solution feed line, and coupling openings exposed through disassembly, shall be sealed with tape conforming to PPP-T-60.

5.1.3 Matchmarking. When necessary to facilitate reassembly, each part removed and the mating parts on the basic unit shall be matchmarked. Matchmarking shall be printed on shipping tags conforming to UU-T-81, Type A. Printed tags shall be waterproofed in accordance with MIL-STD-129.

5.2 Preservation and packaging. The preservation and packaging shall be Level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Scrubbing machine. The scrubbing machine and component parts shall be preserved and packaged in accordance with the applicable provisions of MIL-T-3351 with the following requirements and exceptions.

5.2.1.1.1 Exhaust system. After completion of engine preservation, unpainted exterior surfaces of exhaust pipes, mufflers, flame arresters, etc., shall be coated with type P-1 preservative.

5.2.1.1.2 Brush mechanism components, brushes, rinse system components, pick-up pipe floor nozzle, and similar components removed shall be individually preserved and packaged in accordance with the requirements for such removed items of MIL-T-3351.

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5.2.1.1.3 Water tanks. Interior surfaces of the water tanks shall be preserved by coating with type P-10, grade 2, preservative using either the fill and drain or fogging method of MIL-P-116. Drain plugs and filler cap threads shall be removed, coated with type P-10, grade 2 preservative, and reinstalled.

5.2.1.1.4 Seat backs and seat cushions. Seats and back cushions of scrubbers not equipped with a closed cab shall be covered with waterproofed paper, conforming to UU-P-271, Class E-1 or E-2; all laps, seams, and folds shall be sealed with adhesive conforming to MMM-A-260. The paper shall cover the top, sides, and ends of the seat and back cushions. The cover shall be secured at various points with tape conforming to PPP-T-60, Class I (Type optional).

5.2.1.1.5 Horn buttons. Horn buttons on scrubbers not equipped with a closed cab shall be sealed with tape conforming to PPP-T-60, Type II, Class I.

5.2.2 Level C. Scrubbing machines and component parts shall be preserved and packaged in accordance with the manufacturer's standard practice for commercial scrubbing machines.

5.3 Packing. The packing shall be Level A, B or C, as specified (see 6.2).

5.3.1 Level A. Scrubbing machines and component parts shall be packed in accordance with the requirements of MIL-T-3351, for Level A, Type III, unboxed (mobile) units.

5.3.2 Level B. Scrubbing machines and component parts shall be packed in accordance with the requirements of MIL-T-3351, for Level B, Type III, unboxed (mobile) units.

5.3.3 Level C. Scrubbing machines and component parts shall be packed to insure carrier acceptance and delivery to destination in compliance with ICC Motor Carrier Safety Regulations and other rules and regulations applicable to the mode of transportation.

5.4 Marking. Unless otherwise specified (see 6.2) marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Scrubbers covered by this specification will be used primarily for maintenance cleaning of aircraft ramps, parking areas, and hangar areas where oil and grease drippings, fuel spillage, and dirt tends to collect.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

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b. Type of scrubbing machine required (see 1.2).

c. Special safety devices, if required (see 3.2.1).

d. Painting, if different (see 3.3.3).

e. Nameplate, if different (see 3.6).

f. Identification plate, if not required (see 3.7).

g. Preproduction inspection, if different (see 4.2.1).

h. Quality conformance inspection, if different (see 4.2.2).

i. Level of preservation and packaging required (see 5.2).

j. Level of packing required (see 5.3).

k. Marking, if different (see 5.4).

6.3 Preproduction model. The preproduction model may be a preproduction item or a production item which meets the requirements of this specification. In either case, the model shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a preproduction model shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

Custodians
Army - ME
Air Force - 84

Reviewers
Army - ME
Air Force - 84,85

Users
Army - CE

Preparing Activity
Air Force - 84

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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